Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

Thomas Kwa, Ben West, Joel Becker, et al. — Model Evaluation & Threat Research (METR)

Source: arxiv.org/abs/2503.14499 | March 2025

~7 months
Doubling Time

R^2 = 0.97
Exponential Fit

170 tasks
Benchmark Suite

2 sec → 2 hr
GPT-2 → o3

A New Metric: 50% Task-Completion Time Horizon

1 **Create Diverse Task Suite**

170 software & research tasks ranging from 1 sec to 8 hours of skilled human effort

→

2 **Measure & Compare**

Time skilled humans on each task; measure AI agent success rates across all 170 tasks

→

3 **Fit Logistic Model**

Estimate the task duration at which each AI agent achieves exactly 50% success probability

50% Time Horizon = the task duration (in human-work time) at which an AI agent succeeds 50% of attempts

An intuitive, human-grounded measure enabling apples-to-apples comparison across vastly different models (GPT-2 through o3)

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST 97 tasks
Diverse software tasks
1 min – 30 hrs

RE-Bench 7 tasks
Difficult ML research engineering, all 8 hrs

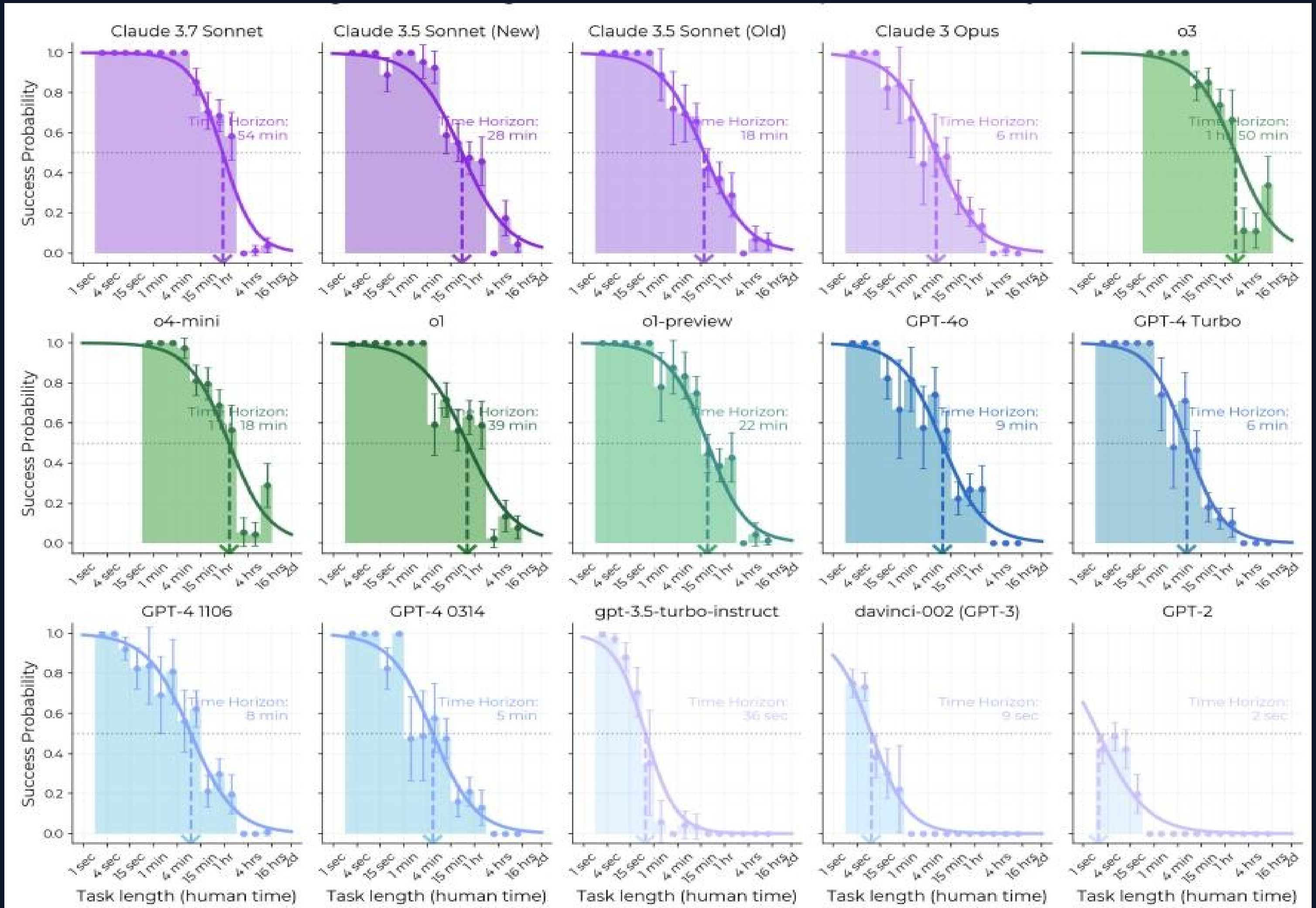
SWAA 66 tasks
Short single-step tasks
1 sec – 30 sec

800+ human baselines
2,529 hours

Task Family	Human Time	Source	Description
find_shell_script	3 seconds	SWAA	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	HCAST	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	HCAST	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	HCAST	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	RE-Bench	Implement custom CUDA kernels for 30x speedup of a backtesting tool

Tasks were selected to span a wide range of difficulty; the suite is not a random sample of real-world work.

AI Task Horizon: 2 Seconds (GPT-2) → ~2 Hours (o3) in 6 Years



Doubling Time

~207 days (~7 months)

Goodness of Fit

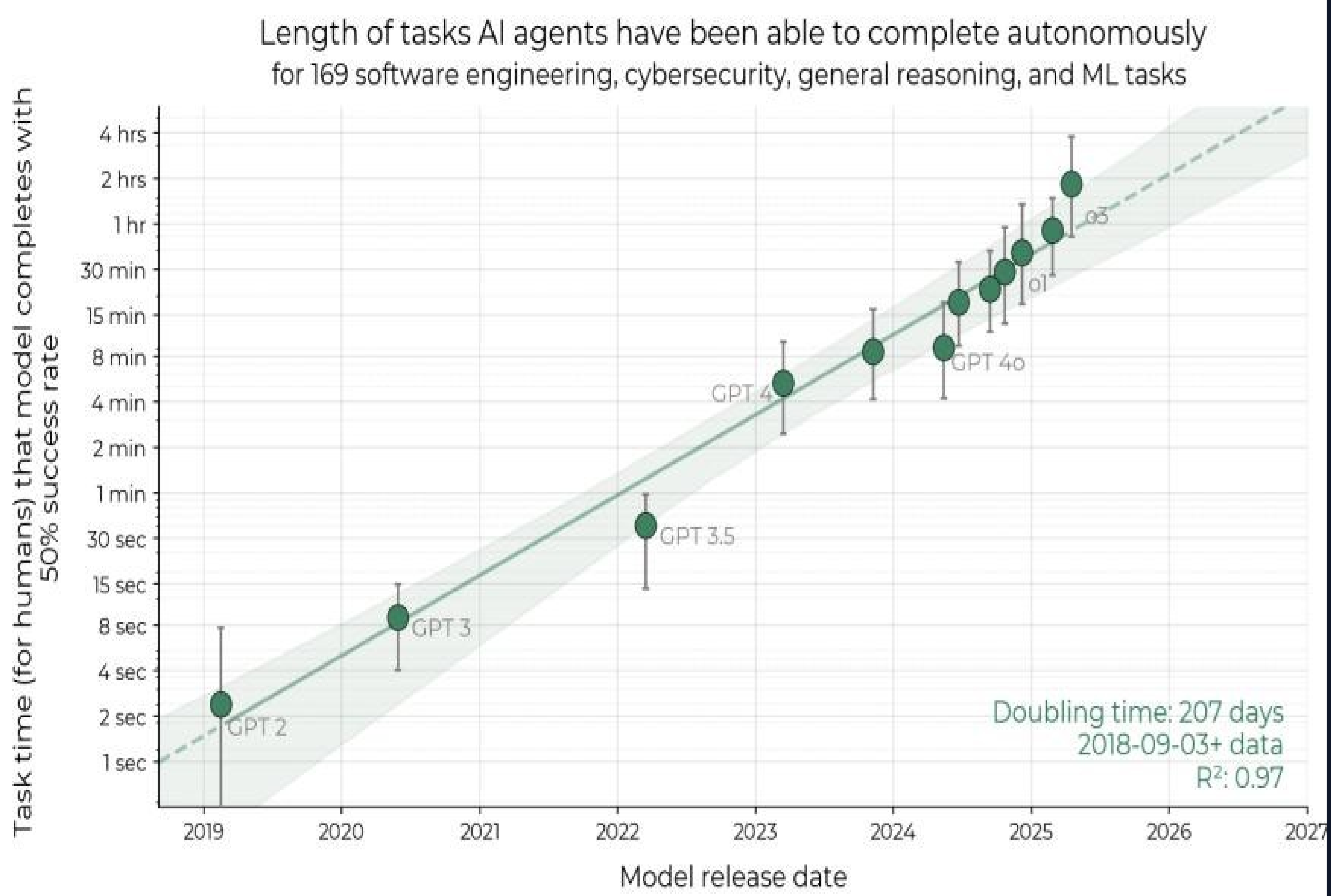
$R^2 = 0.97$

Key Milestones

GPT-2 (2019) **~2 seconds**
GPT-4 (2023) **~5 minutes**
Claude 3.5 Sonnet **~28 minutes**
o1 (2024) **~39 minutes**
Claude 3.7 Sonnet **~54 minutes**
o3 (2025) **~1 hr 50 min**

2023–2025 rate is ~20% faster than overall trend

Success Probability Drops Sharply as Task Length Increases



Reading the Curves

Each curve shows one model's success rate vs. human task duration. Newer models push the curve rightward — succeeding on longer tasks.

Frontier Comparison

GPT-2 (2019) ~2 sec

Claude 3.7 Sonnet ~54 min

o3 (2025) ~1 hr 50 min

80% Time Horizon

Tasks completed 80% of the time show a similar trend but with horizons roughly 5× shorter.

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

01 Improved Logical Reasoning

Models increasingly handle multi-step deduction chains, producing correct solutions for tasks requiring sustained logical thought.

02 Better Tool Use

Modern agents wield code execution, file systems, web browsing, and APIs with growing fluency — extending reach beyond pure text generation.

03 Reliability & Self-Awareness

Agents detect when they are stuck or confused, ask clarifying questions, and re-plan — reducing cascading failures on long tasks.

04 Adaptation to Mistakes

Rather than halting after an error, frontier models debug, backtrack, and retry alternative approaches — key for multi-hour tasks.

Key Limitation

Performance is notably lower on less structured, "messier" tasks. External validity remains an open question — tasks may not perfectly represent average real-world intellectual labor, though supplementary experiments found little evidence that trends are slower on more realistic tasks.

Extrapolation Predicts Month-Long Task Automation by 2028–2031

Forward Extrapolation

If the ~7-month doubling continues:

- 1 day of human work → ~2026–2027
- 1 week of human work → ~2027–2029
- 1 month (167 hrs) → mid-2028 – mid-2031

The 2023–2025 growth rate is ~20% faster than the overall 2019–2025 rate, suggesting possible acceleration.

Caveats & Uncertainties

- Exponential trends rarely continue indefinitely — structural ceilings (data, compute, task complexity) may slow growth.
- External validity is uncertain: the 170-task suite does not perfectly represent average real-world intellectual labor.
- Supplementary experiments found little evidence that trends are slower on more realistic tasks — but sample sizes are limited.
- Messy, unstructured tasks remain significantly harder; the gap may widen or narrow unpredictably.

Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1

Robust Metric

50% time horizon provides an intuitive, human-grounded metric for tracking AI autonomy — $R^2 = 0.97$ over 6 years of releases.

2

Rapid Doubling

Capability doubles every ~7 months, with possible acceleration since 2024. This pace demands continuous monitoring frameworks.

3

Current Frontier

o3 autonomously completes tasks taking skilled humans nearly 2 hours — up from 2 seconds for GPT-2 in 2019.

4

External Validity Gap

The 170-task suite may not perfectly represent real-world intellectual labor. Supplementary experiments are encouraging but limited in scope.

5

Unstructured Task Limitation

Performance remains notably lower on messy, less-structured tasks — uncertain generalization to real-world job distributions.

Source: Kwa et al. (METR), "Measuring AI Ability to Complete Long Software Tasks," arXiv:2503.14499, March 2025